



A Desktop Twitter Client in 15 Minutes

Twitter, the fun social-networking micro-blogging service, has taken the world by storm. Millions of 'tweets'—real-time text messages of up to 140 characters—are posted daily, using several different client applications. Users can read other users' tweets via the Twitter website, mobile Short Message Service (SMS), Really Simple Syndication (RSS) feeds, and Twitter clients. Twitter clients use a published Twitter API to interact with Twitter servers; this tutorial shows how you can build your own Twitter application to tweet, or search tweets, via the Twitter API.

Twitter is known not only for the simplicity of its concept, but also for its super-cool and easy-to-use API. So let's quickly try and develop our own desktop Twitter client in three simple steps, producing both a command-line and GUI interface for the Twitter client.

Step 1: Command-line tweeting

Twitter's simple API allows you to post messages from the command line using cURL (<http://curl.haxx.se/>). If it's not installed on your system, use your distribution's package manager to install it—packages are available for practically every major distro.

An example invocation:

```
curl --basic --user <username>:<password> --data
status=<message> http://twitter.com/statuses/update.xml
```

Let's examine the parts of the command:

--basic <username>:<password> lets you supply your Twitter username and password, for the basic authentication required by the API.

--data status=<message> will send a POST HTTP request containing your new message, to the URL <http://twitter.com/statuses/update.xml>. This XML file will be used to send the messages to the user's account.

To carry out a search via the command line, use a command like the following:

```
curl http://search.twitter.com/search.atom?q=worldcup
```

This command will search Twitter for the string 'worldcup', and will return XML from the search page, containing the results.

Step 2: Java program for Twitter

There is a simple function in Java that allows us to execute a Linux command (who cares about Windows!) from within a user program, and retrieve the results, provided the command is installed on the system. Let's see how the function can be used to execute our very own Linux command to send a tweet:

```
String s="curl --basic --user"<username>:"<password>"--data
status="+ message +"http://twitter.com/statuses/update.xml";
```